

Feetech STS3215 Servo

Register Address Reference

For use with scservo_sdk Python library

EEPROM Registers (Non-Volatile)

These values are saved even after power off. Torque must be disabled to write.

Addr	Name	Size	R/W	Default	Description
0	Firmware Major	1	R	-	Major firmware version number
1	Firmware Minor	1	R	-	Minor firmware version number
3	Servo Major	1	R	-	Servo series major version
4	Servo Minor	1	R	-	Servo series minor version
5	ID	1	R/W	1	Servo ID (0-253), must be unique on bus
6	Baud Rate	1	R/W	0	0=1Mbps, 1=500K, 2=250K, 3=128K, 4=115200, 5=76800, 6=57600, 7=38400
7	Return Delay	1	R/W	0	Response delay time in microseconds
8	Response Level	1	R/W	1	0=No response, 1=Respond to read, 2=Respond to all
9	Min Angle Limit	2	R/W	0	Minimum position limit (0-4095); set both to 0 for motor mode
11	Max Angle Limit	2	R/W	4095	Maximum position limit (0-4095); set both to 0 for motor mode
13	Max Temp Limit	1	R/W	70	Maximum temperature limit in °C
14	Max Voltage Limit	1	R/W	140	Maximum voltage limit (value × 0.1V)
15	Min Voltage Limit	1	R/W	50	Minimum voltage limit (value × 0.1V)
16	Max Torque Limit	2	R/W	1000	Maximum torque output (0-1000 = 0-100%)
18	Phase	1	R/W	0	Motor phase/direction setting
19	Unload Condition	1	R/W	0	Conditions that trigger torque release
20	LED Alarm	1	R/W	0	LED blink conditions for alarms (bitmask)
21	P Coefficient	1	R/W	32	PID proportional gain
22	D Coefficient	1	R/W	32	PID derivative gain
23	I Coefficient	1	R/W	0	PID integral gain
24	Min Startup Force	2	R/W	0	Minimum torque to start moving (0-1000)
26	CW Dead Zone	1	R/W	1	Clockwise dead zone width
27	CCW Dead Zone	1	R/W	1	Counter-clockwise dead zone width

28	Protection Current	2	R/W	0	Overcurrent protection threshold ($\times 6.5\text{mA}$)
30	Angular Resolution	1	R/W	1	Position resolution multiplier
31	Position Offset	2	R/W	0	Position offset calibration value
33	Mode	1	R/W	0	0=Position, 1=Constant speed, 2=PWM, 3=Step servo
34	Protection Torque	1	R/W	20	Torque during overload protection (%)
35	Protection Time	1	R/W	200	Time before overload protection triggers
36	Overload Torque	1	R/W	80	Torque threshold for overload detection (%)

RAM Registers (Volatile)

These values reset on power cycle. Used for real-time control and status.

Addr	Name	Size	R/W	Description
40	Torque Enable	1	R/W	0=Torque off (servo is limp), 1=Torque on (servo holds position)
41	Acceleration	1	R/W	Acceleration for movement (0=instant, 1-254=slower to faster ramp)
42	Goal Position	2	R/W	Target position (0-4095); in motor mode this is speed/direction
44	Goal Time	2	R/W	Time to reach goal position in milliseconds
46	Goal Speed	2	R/W	Speed limit for movement (0=max speed, 1-4095=limited)
48	Torque Limit	2	R/W	Current torque limit (0-1000 = 0-100%)
55	Lock	1	R/W	0=EEPROM unlocked, 1=EEPROM locked (write protected)
56	Present Position	2	R	Current actual position (0-4095)
58	Present Speed	2	R	Current movement speed
60	Present Load	2	R	Current load/torque (bits 0-9=magnitude, bit 10=direction)
62	Present Voltage	1	R	Current voltage (value $\times 0.1\text{V}$)
63	Present Temperature	1	R	Current temperature in $^{\circ}\text{C}$
64	Async Write Flag	1	R	Status of asynchronous write operation
65	Servo Status	1	R	Error status bitmask (voltage, temp, overload, etc.)
66	Moving	1	R	1=Servo is currently moving, 0=Stationary
69	Present Current	2	R	Current draw (value $\times 6.5\text{mA}$)

Quick Reference - Common Operations

Operation	Register(s)	Example Value	Notes
Enable torque	40	1	Must enable before servo responds to goal positions

Disable torque	40	0	Servo goes limp; can move by hand, still read position
Move to position	42	0-4095	2048 = center position
Set speed limit	46	0-4095	0 = maximum speed
Read position	56	-	Returns current position 0-4095
Read load	60	-	Returns load magnitude and direction
Set acceleration	41	0-254	0 = instant, higher = smoother ramp
Motor mode	9 & 11	both 0	Set min and max angle limits to 0
Servo mode	9 & 11	0 & 4095	Restore angle limits for position control

Notes:

- **Size:** Number of bytes (1 or 2). Use read1ByteTxRx/write1ByteTxRx for 1-byte, read2ByteTxRx/write2ByteTxRx for 2-byte registers.
- **Position range:** 0-4095 represents 0-360°. Each unit is approximately 0.088°.
- **Default baud rate:** 1,000,000 bps (1 Mbps).
- **Default servo ID:** 1.
- **Voltage values:** Multiply by 0.1 to get volts (e.g., 74 = 7.4V).
- **Current values:** Multiply by 6.5 to get milliamps.
- **Torque/Load values:** 1000 = 100% of maximum.
- **EEPROM writes:** Disable torque first, then write, then re-enable torque.